

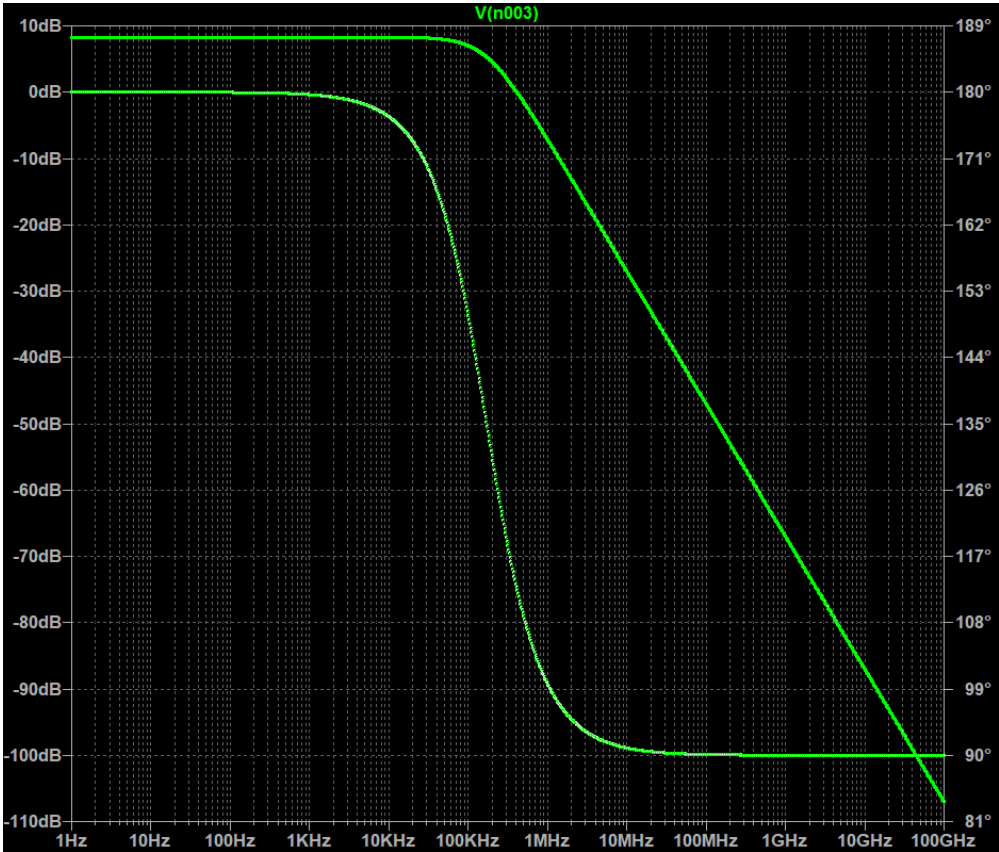
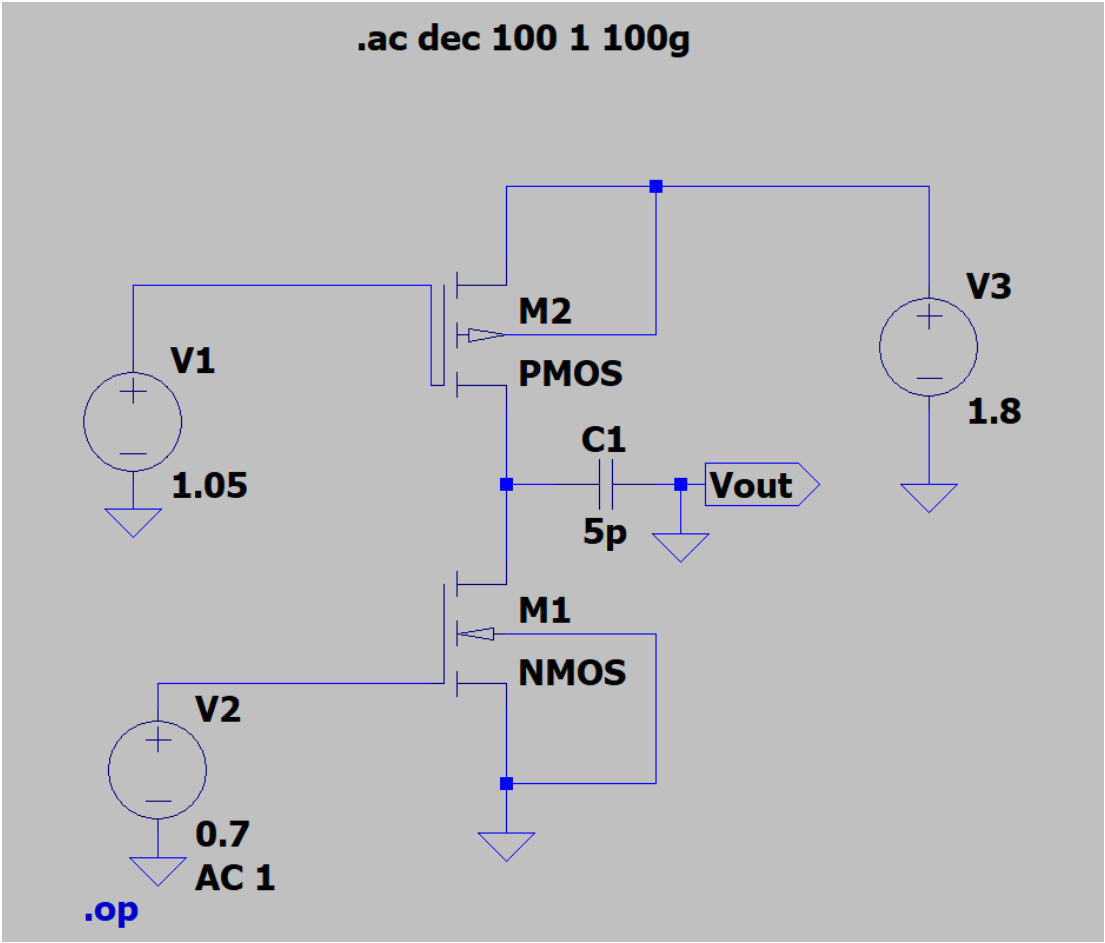
ASSIGNMENT 03

FAROOQ ABDULLA

NIELIT Online Internship on Analog and Digital VLSI Design Flow

Batch 1

Q1) Design a PMOS load common source amplifier and calculate gain and pole location (a) Verify with the gain vs frequency plot (b) Mark Unity Gain Frequency and Find Phase Margin Attach the plots and Tabulate the results.

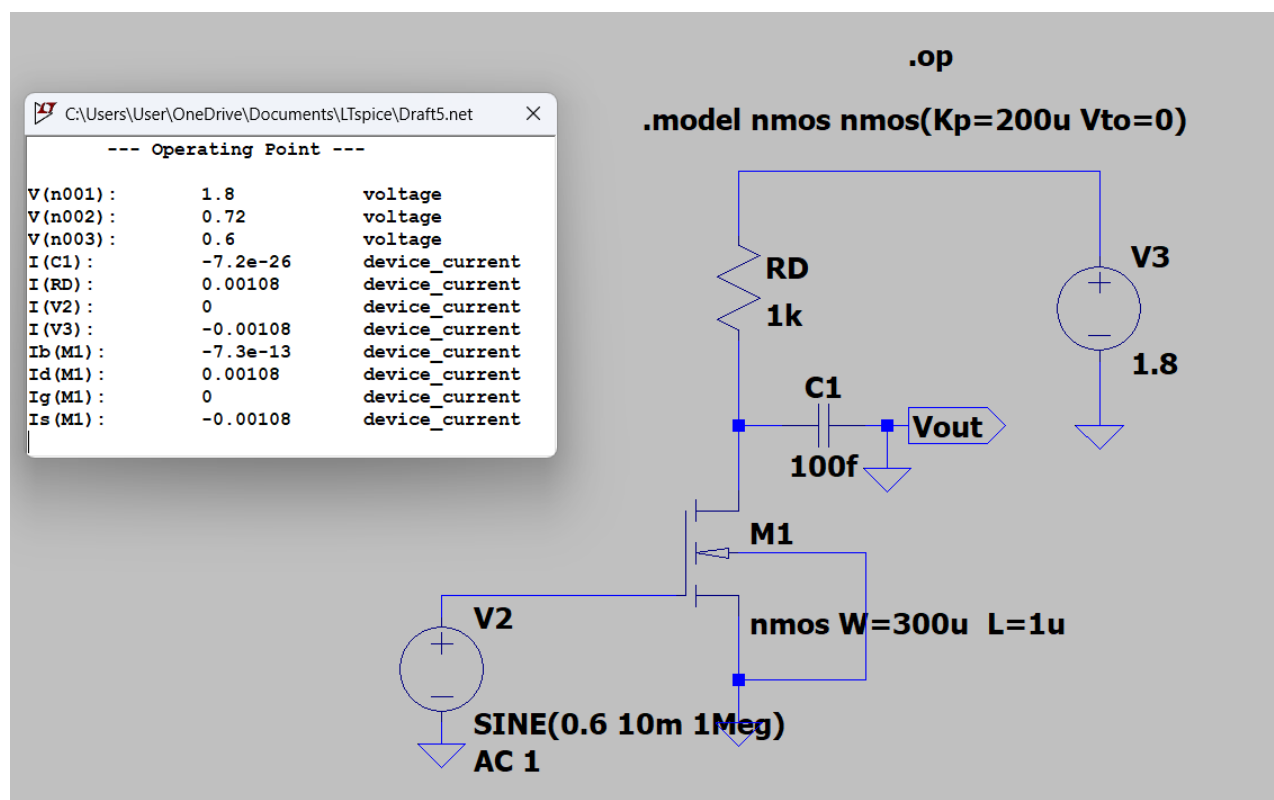


Parameter	Value Extracted	Visual Reference
DC / Midband Gain (A_v)	8.3 dB	Flat low-frequency region (1 Hz to 10 kHz)
Dominant Pole (f_p)	300 kHz	Frequency where Phase = 135° / Gain = 5.3 dB
Unity Gain Frequency (UGF)	750 kHz	Frequency where Magnitude curve crosses 0 dB
Phase at UGF (ϕ)	90°	Read from the phase axis at 750 kHz
Phase Margin (PM)	90°	Excellent stability

Q2) Design a resistive load common source amplifier to obtain a gain of -5 V/V. The amplifier is expected to drive a load capacitance of 100fF.

- Perform DC, AC and transient simulations to verify the gain
- From the AC analysis, observe the gain, bandwidth, gain-bandwidth product and poles/zeros of the designed amplifier
- Perform noise analysis and find input referred voltage noise of your circuit

DC Simulation

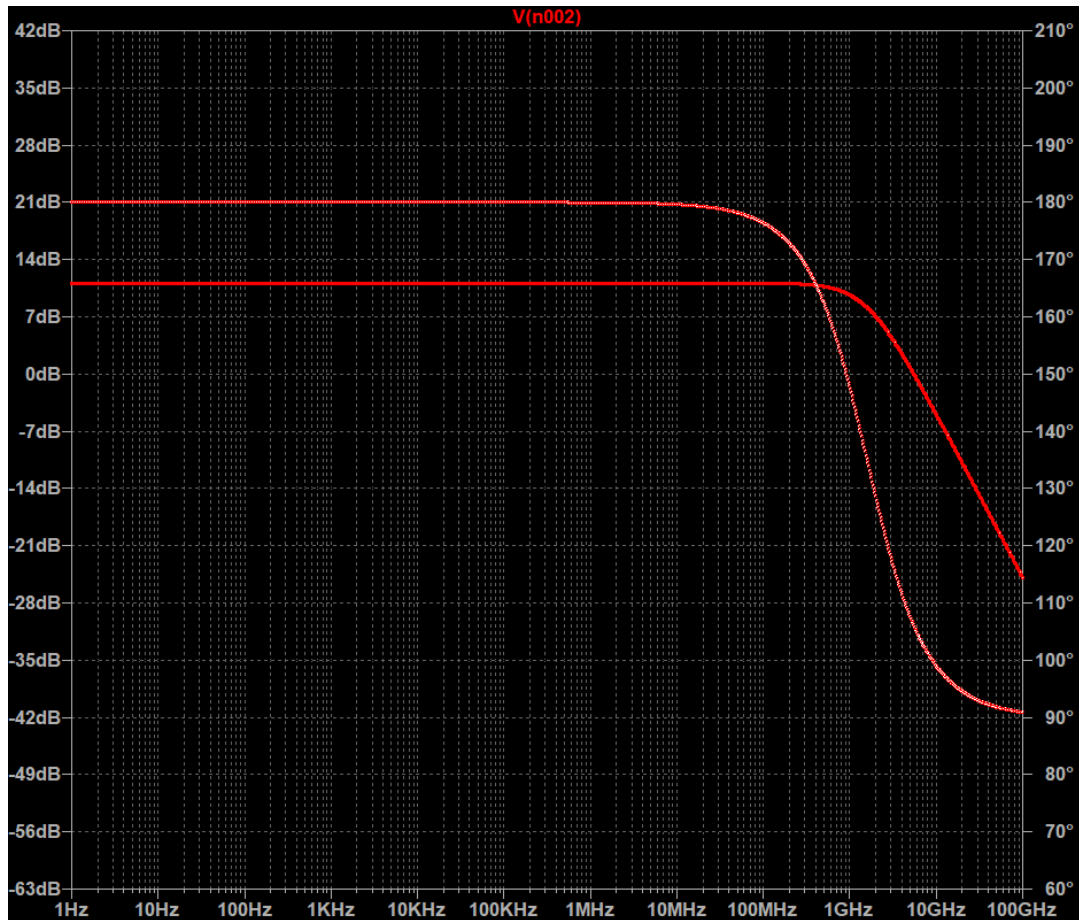
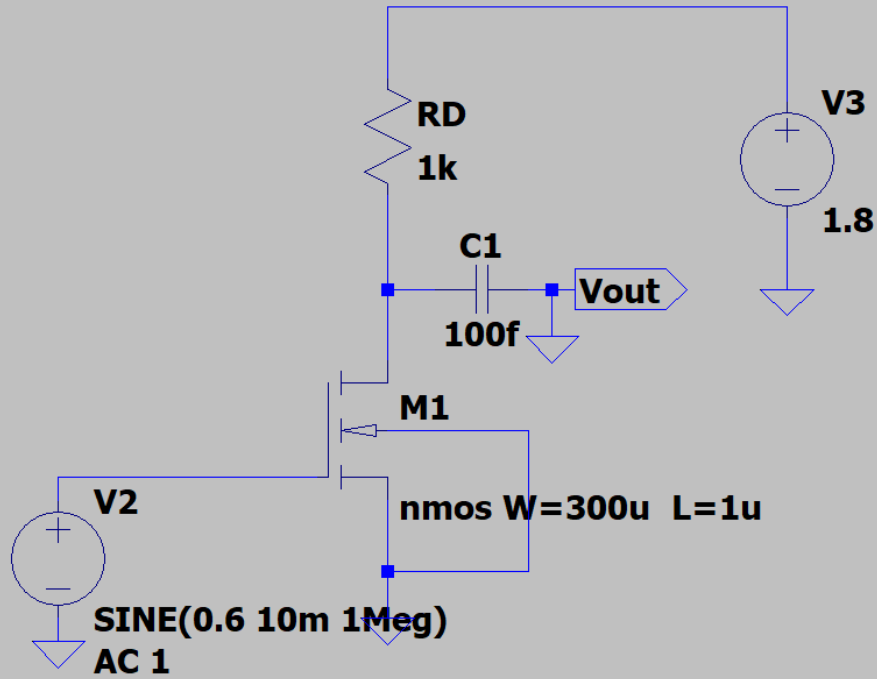


AC simulation :

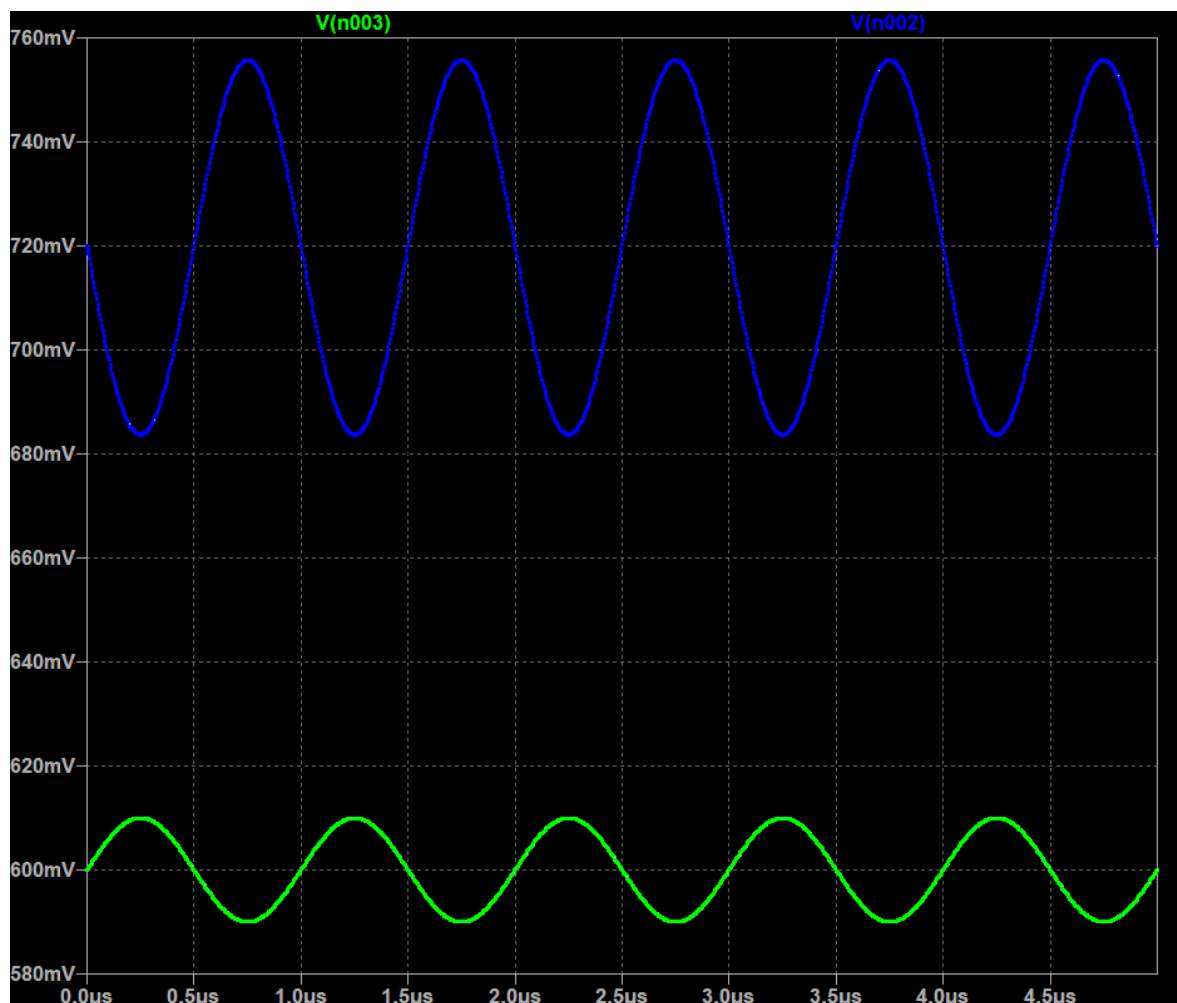
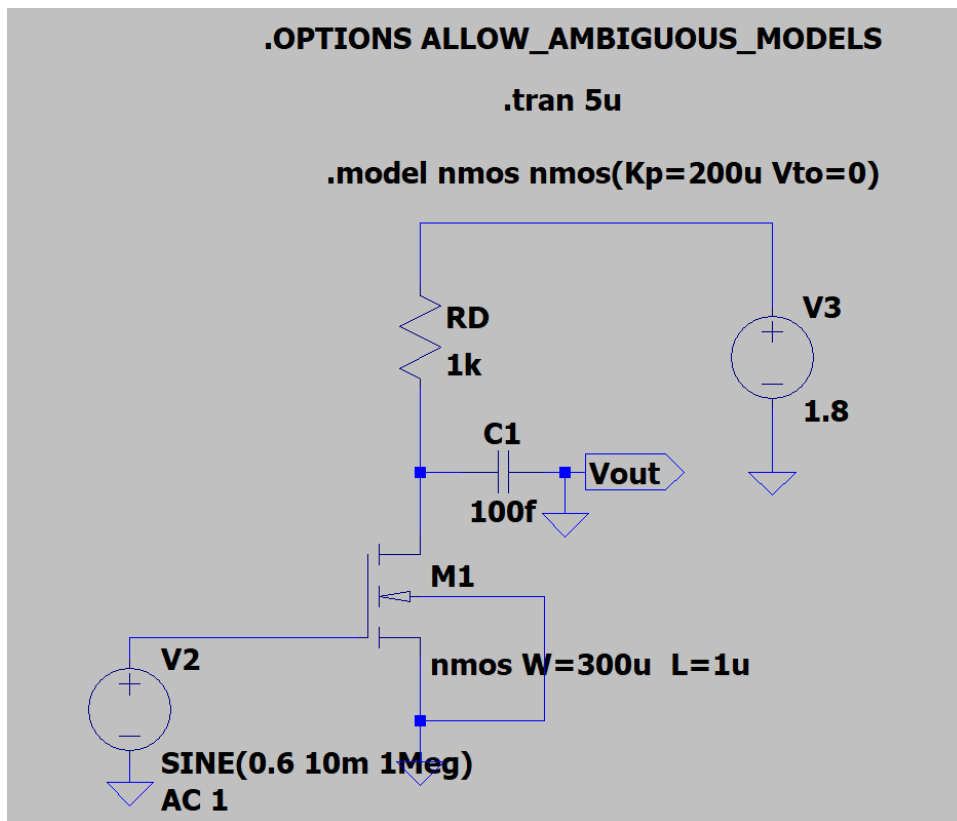
.OPTIONS ALLOW_AMBIGUOUS_MODELS

.ac dec 100 1 100g

.model nmos nmos(Kp=200u Vto=0)

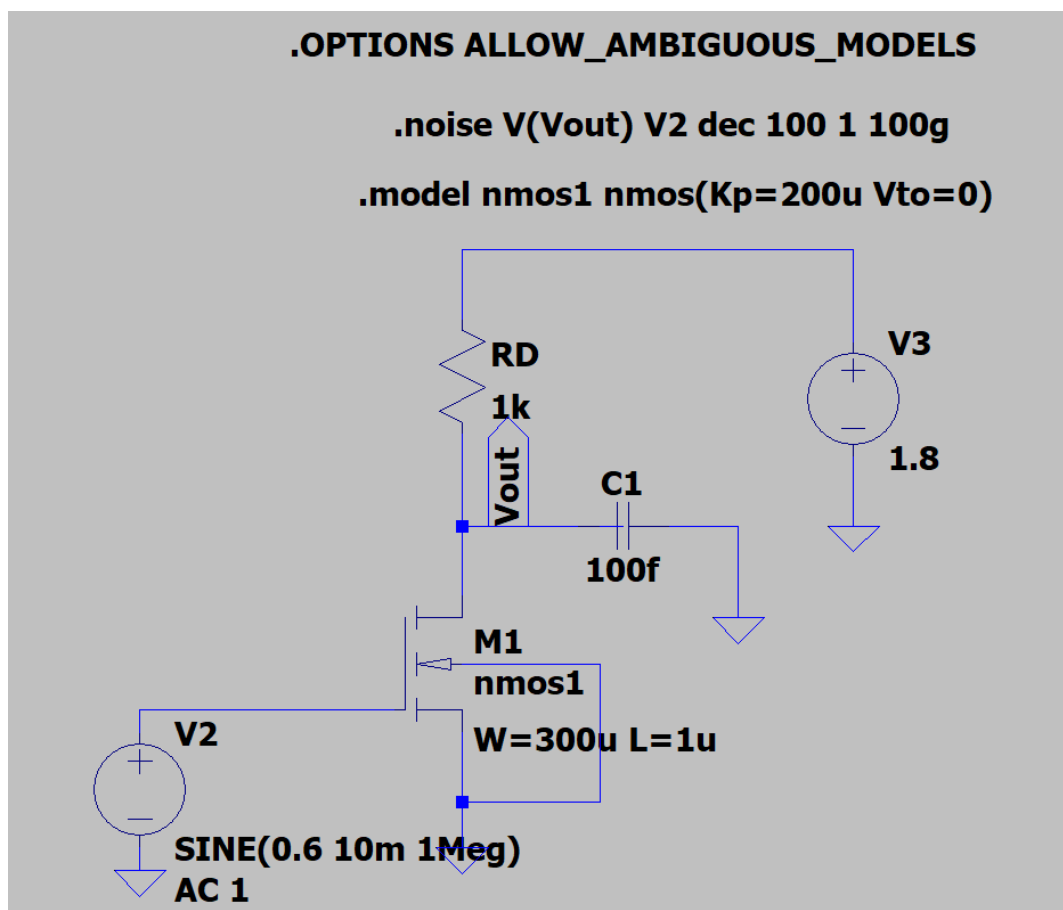


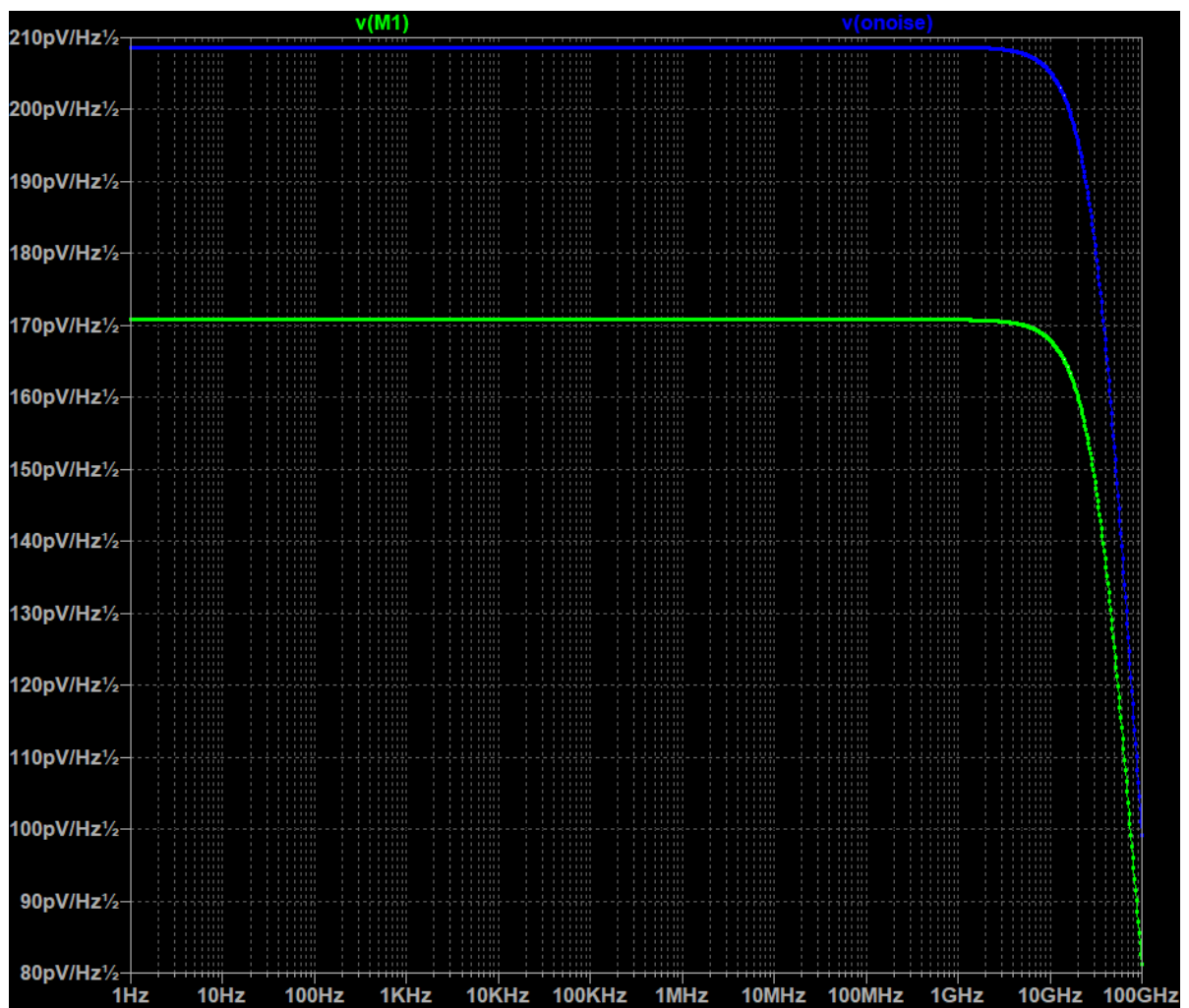
Transient Simulation:



Parameter	Value (Observed / Calculate)
Low-Frequency Gain (dB)	~11.5 dB
Low-Frequency Gain (Linear)	~3.76 V/V
Bandwidth (–3 dB Frequency)	~2 GHz
Gain-Bandwidth Product (GBW)	~7.52 GHz
Dominant Pole	~2 GHz
Zeros	None

Noise analysis:





input referred voltage noise of your circuit = $\sim 41.6 \text{ pV}/\sqrt{\text{Hz}}$